Balance Rehabilitation

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PT4000-YEAR|PT7003-YEAR

Physiotherapy Theory and Practice 1: Long Term conditions, Physiotherapy Theory & Practice 1

November 19, 2024 9:00 - 13:00

Lecture Physical

469 154

Learning Outcomes

After this lecture, students should be able to:

- Identify client groups that may benefit from balance rehabilitation
- Understand the components that may be included as part of a balance rehabilitation exercise plan
- Understand the parameters that can be used to progress balance exercises

General Information about balance disorders

- According to the American Medical Association, 1 in every 4 patients suffers from some level of balance dysfunction.
- 40% of over 65 years old have balance disorders sufficient to put them at risk of falling
- Approximately 2 million physician visits per year involve chief complaints of dizziness or loss of balance
- In 2001 alone, 1.6 million older adults were treated in emergency rooms for fall related injuries

Client groups which may benefit from balance Rehabilitation

- Older people who fall
- Patients with peripheral or central vestibular disorders
- Patients with neurological problems (e.g. stroke, multiple sclerosis, Parkinson's, traumatic brain injury), cerebral palsy, motor neuron disease, functional neurological dysfunction etc
- Patients with chronic musculoskeletal problems (chronic ankle sprains)

Impairments associated with balance dysfunction

- ↓ muscle strength
- ↓ muscle power
- ↓ muscle endurance
- ↓ coordination
- Disordered sensory and perceptual processes
- Cognitive dysfunction
- Executive function impairment

- Adaptive changes in muscle length following immobility and disuse – reduced range of motion (ROM)
- Impaired motor control
- Visual, tactile, proprioceptive and vestibular impairments
- Damage of higher centres e.g. Basal ganglia

These impairments contribute to ↑ incidence of falls

Factor	Impairments	Strength of association
Vision	Poor visual contrast sensitivity	***
	Decreased depth perception	***
	Poor visual acuity	**
Peripheral sensation	Reduced vibration sense	***
	Reduced tactile sensitivity	***
	Reduced proprioception	**
Muscle Strength	Reduced muscle strength	***
	Reduced muscle power	*
	Reduced muscle endurance	*
Reaction Time	Poor simple reaction time	***
	Poor choice reaction time	***

Sensory and neuromuscular factors associated with falls in the elderly

Components of Balance Rehabilitation

Goals of task-oriented balance rehabilitation

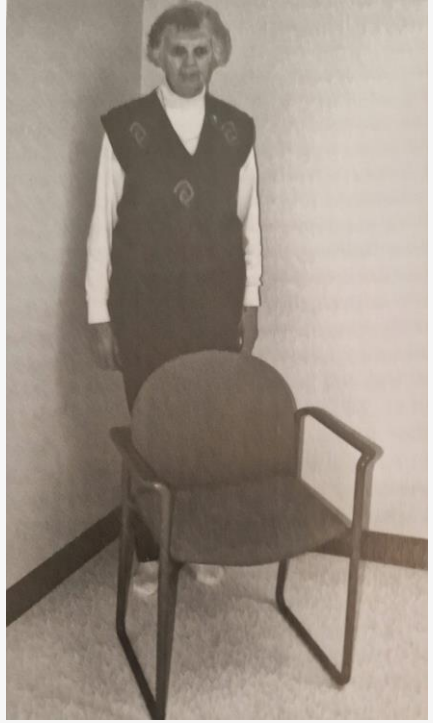
1. **Impairment level:** Resolve, reduce and prevent impairments in system that are important to balance e.g. musculoskeletal system.

2. **Strategy Level:** Develop effective task-specific sensory, motor and cognitive strategies

3. **Functional Task Level:** Retrain functional tasks with varying postural control demands e.g. static, reactive, anticipatory control under different environmental conditions.











Patient Safety

- At home and in gym
- Parallel Bars
- Gait belts
- Guarding patients or providing instruction on safe position when practicing exercises independently

1. Impairment Level: musculoskeletal impairments

1

Improving muscle strength

2

Improving muscle length

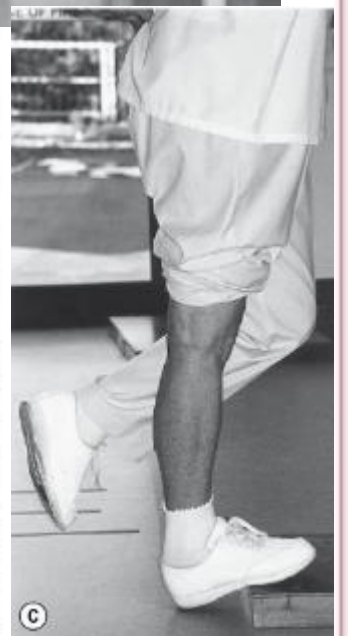
Examples of approaches to improve muscle strength











Box 2.1 Guidelines for functional strength training

- Exercises should be task-oriented – step-ups, squats, push-ups – and specific to tasks being learned plus standing up/sitting down, walking, stair climbing/descent, reaching to targets
- Encourage patient to exercise as intensively as possible, to the point of mild muscle fatigue*
- Grade the amount of resistance and number of repetitions to the individual's ability
- Utilize resistance from body weight, free weights, elastic bands, isokinetic dynamometry, exercise machines, inclined treadmill, depending on specific effects required and the existing muscle control and strength
- Vary body weight resistance by changing step height, chair height, weight of object being lifted
- As a general rule, a maximum number of repetitions should be performed (<10) without a rest and repeated in three sets, with a short break between sets
- In progressive resistance weight training, the person attempts between 3 and 20 repetitions of 50% of the maximum possible load that can be lifted in one attempt. The same principle for repetitions is followed for task-specific exercises, but using body weight for load
- For endurance, a higher number of repetitions is practised at low levels of load and exercises include stationary cycling, arm cycling and treadmill walking
- For very weak muscles, use methods which facilitate muscle activation and force generation, and include simple exercises, biofeedback, mental practice, functional electrical stimulation.
- A normal breathing pattern should be maintained throughout exercise

*Individuals at risk from high levels of blood pressure should avoid high-intensity resistance training.
Refer to American College of Sports Medicine's Guidelines for Exercise Testing and Prescription 2006.

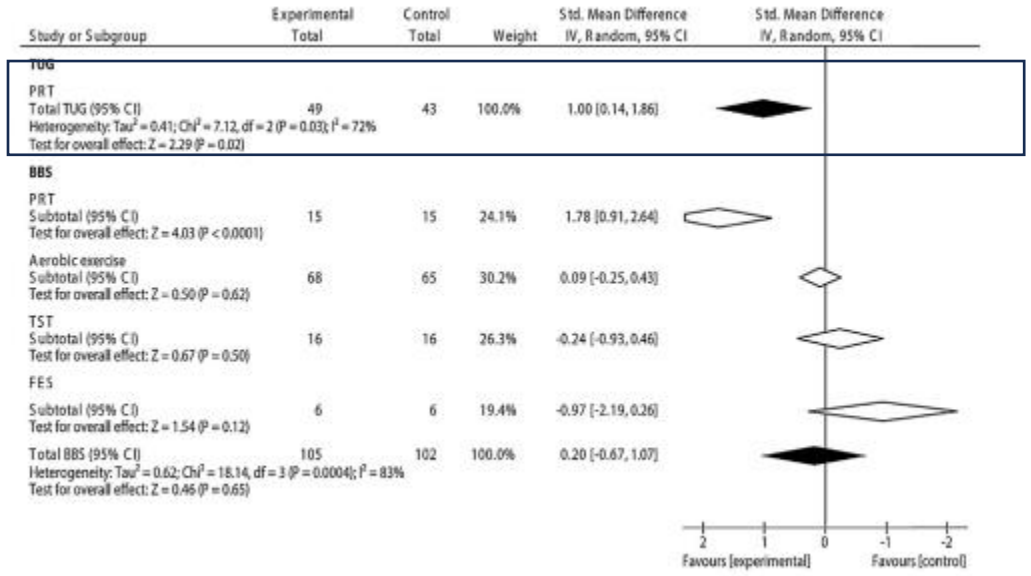
Evidence for muscle strengthening

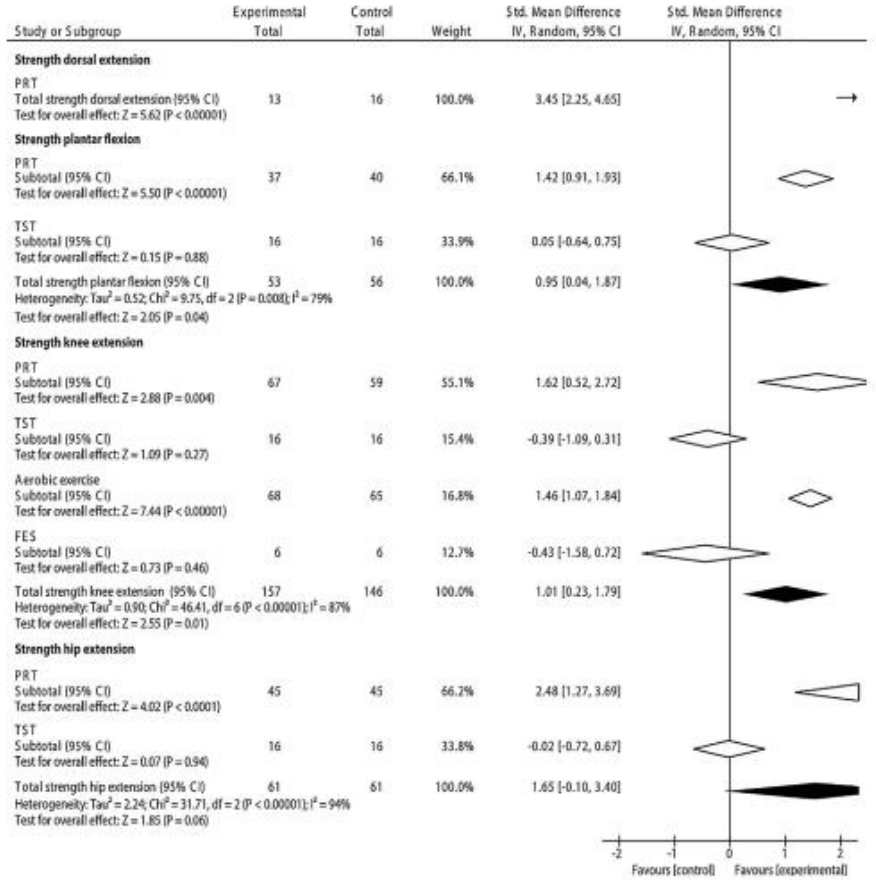
- Many studies have shown that **resistance training is effective** in **improving strength in older adults,** and in some instances associated with improvement in balance and not in other studies .

- Systematic review (Orr et al. 2008) of RCTs to evaluate the effect of progressive resistance training on balance
  - 29 studies
  - 14 found improvements in balance
  - Inconsistent results suggest that resistance training alone is not sufficient to improve balance control in older adults

- Similar mixed results in neurological population
  - Improvement in power but no improvement in postural control (anterior posterior sway) (Debolt & McCubbin 2004)

Muscle strengthening for hemiparesis after stroke: A meta-analysis



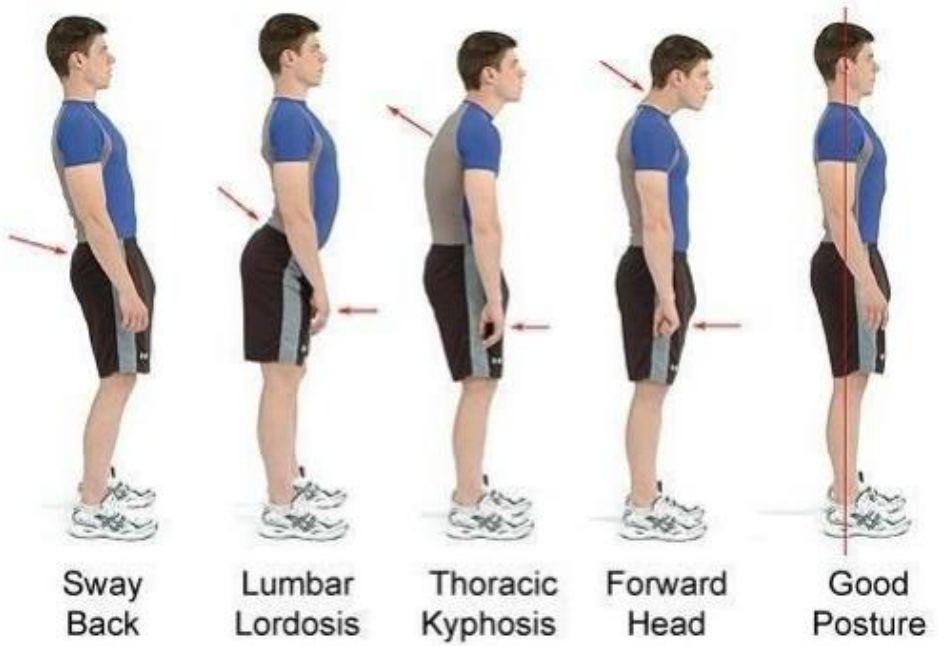


Sophie W et al. 2016 https://pubmed.ncbi.nlm.nih.gov/26969343/

2: Strategy Level:

- Goal:
  - Facilitate development of effective strategies for meeting postural control demands in a variety of functional tasks requiring static, anticipatory and reactive balance.
- Approach:
  - Progressively challenging tasks
  - Utilise motor learning principles
  - Static – supported - to sitting unsupported – anticipatory – reactive balance - unstable surfaces etc

- These include strategies for:
- Postural strategies
- Movement strategies
- Sensory strategies
- Cognitive strategies



Postural strategies: Retraining postural alignment

- Goal: to help patient develop an initial position that is:
  - Appropriate
  - Efficient with respect to gravity
  - Maximizes stability that places the line of gravity within a patients limits of stability





- Verbal and manual cues, mirrors, and kinetic or force feedback devices have been used to retrain vertical posture
- Patients practice with eyes open and closed – learning to maintain symmetrical posture without visual cues.

Evidence on use of feedback on symmetrical posture?

- Barclay –Goddard et al. (2004) – systematic review examined the use of force-plate biofeedback to improve standing balance
  - Found that force biofeedback resulted in patients standing more symmetrically but did **NOT** result in improved balance during functional activities.

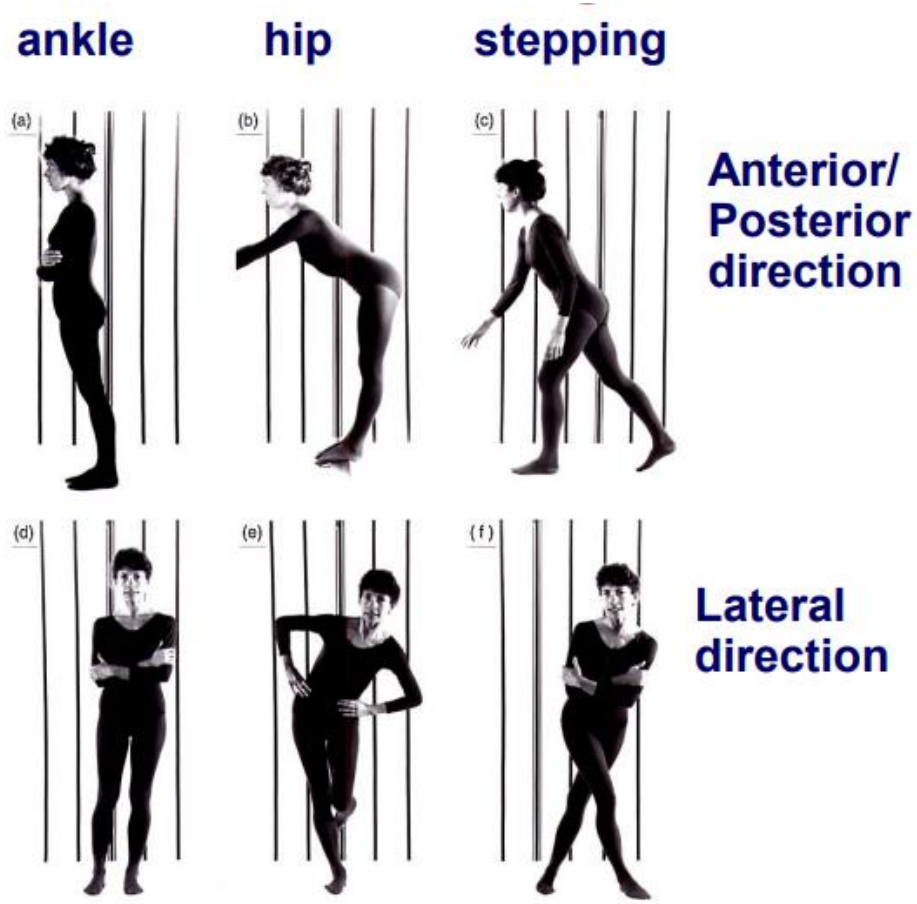
- Van Peppen et al. (2006) – systematic review
  - found a non-significant effect in favour of visual feedback on weight distribution, postural sway, balance, and gait speed in stroke patients
  - Concluded there is no difference between visual feedback and conventional therapy.

Movement Strategies

- Goal: To help the patient develop co-ordinated multi-joint movement that are effective in meeting the demands for postural control in sitting and in standing

- Includes:
  - Strategies that move the COG relative to a stationary base of support (in-place strategies)
  - Strategies that change the base of support relative to COG (changing base of support strategies)
  - Strategies that train anticipatory balance control

Retraining Reactive Balance Strategies



- **Approach**
  - Use external perturbations (direction, speed and amplitude)
  - Small perturbations (in –place)
  - Larger perturbations (stepping)
  - Mixed use of perturbations
  - Multiple directions
  - Eyes open, closed
  - Compliant and inclined surfaces
  - Cognitive conditions

Research Evidence on re-training movement strategies

- There is evidence that training of reactive balance in neurological patients improve the organisation, and timing of postural responses
  - Marigold et al (2005) found improvements in the onset latency of paretic lower extremity muscles in response to platform perturbations in people with chronic stroke after 10 weeks of training.
  - Gatts & Woollacott (2006 & 2007) found that three weeks of intense Tai-Chi training significantly enhanced balance responses by improving the organisation and timing of stepping strategies of the swing leg.

Retraining Anticipatory Balance Control

- Patients with anticipatory balance control problems often are very unsteady or extremely fearful of falling

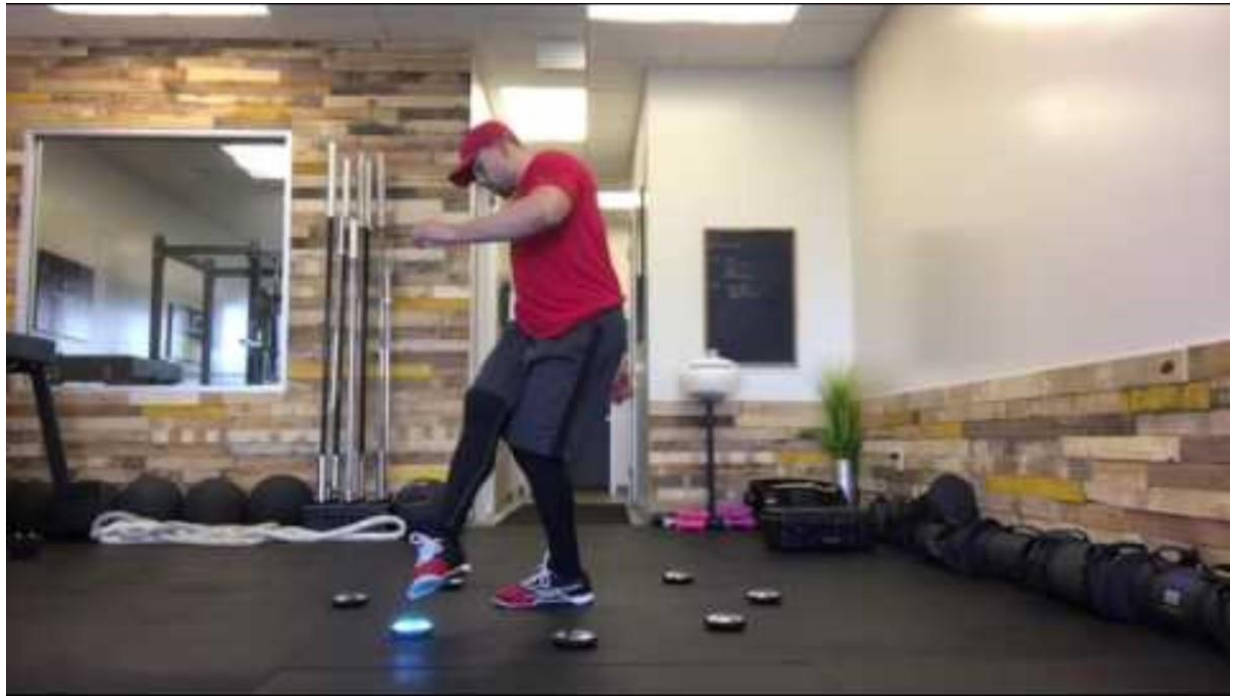
- Indication for anticipatory postural adjustment
  - Inability to move to perform a task
  - Inability to shift weight

Approach

- Patient can practice controlling centre of gravity in all directions, over larger areas at varying speed.
- Example of movement include reaching, lifting, and throwing.
- Progression can be achieved by increasing:
  - Task demands
  - Speed, effort, degree of external support and task complexity
- Patients can practice movement in:
  - Parallel bars
  - Standing close to a wall
  - In a corner with a chair or table in front
  - Therapist can modify the home environment for continuing practice outside of therapy time.

Anticipatory postural adjustment tasks





Research Evidence for anticipatory postural adjustments

- Garland et al. 2003
  - Used surface EMG to investigate the timing of muscle activity in the lower extremity relative to the initiation of arm movement in an arm raising task in patient with stroke
  - Before rehabilitation, muscles of the paretic lower limb were activated after the initiation of the arm movement rather than before.
  - After 1 month rehabilitation, arm movements were **faster,** and recruitment of leg muscles faster.
  - 10 out 27 patients showed activation of anticipatory muscle activities in the hemiparetic limb
  - 12 showed no change in the timing of muscle activity

Sensory strategy

- **Goal:** help patient learn to effectively co-ordinate and select appropriate sensory information for balance
- **Approach:** Patient required to maintain balance during progressively more difficult static and dynamic movement tasks while the clinician systematically vary the availability and accuracy of one or more senses for orientation

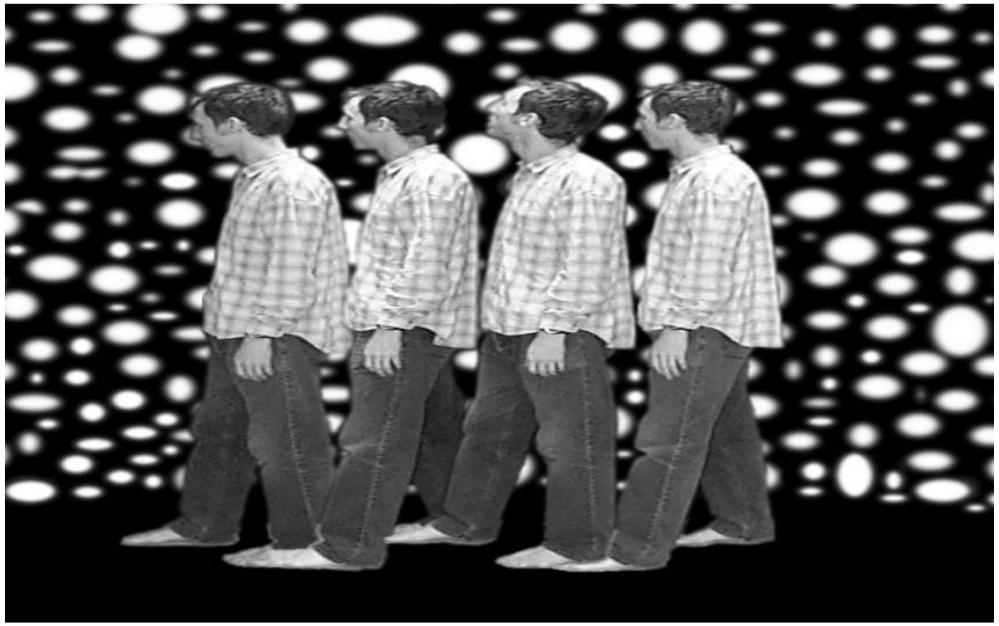
- **Surface dependent**
  - Perform tasks while sitting or standing on surfaces with decreased somatosensory cues for balance:
  - Carpet, foam, pillows, moving surfaces wobble boards.
- **Visual dependent**
  - Perform tasks where visual cues are absent (eyes closed, blindfolded) or reduced (blinders, diminished lighting).
  - Visual cues can also be made inaccurate through use of glasses smeared with petroleum jelly, prism glasses, moving curtains with stripes, moving visual scenes.
- **Enhance ability to use vestibular information**
  - Exercises where one must maintain balance while both visual and somatosensory inputs are simultaneously reduced, such as standing on foam or a wobble board with eyes closed or moving visual scene

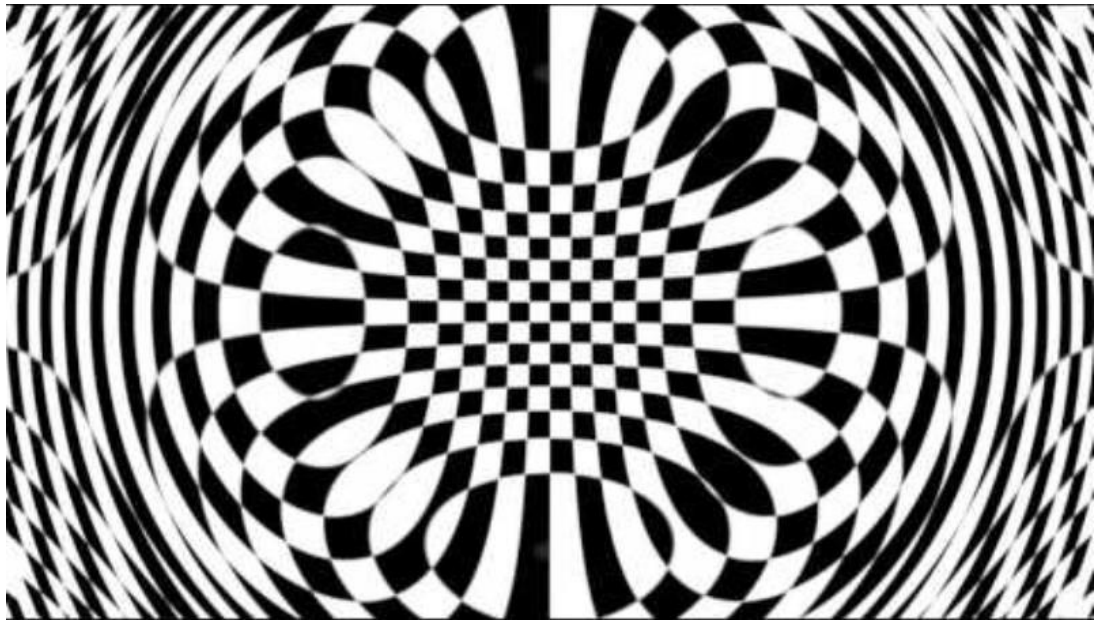
Exercises for training sensory strategies





Examples of optokinetic stimulation





Research Evidence

- A number of studies have demonstrated improvements in sensory organisation for balance through training of:
  - older adults (Hu & Woollacott 1994)
  - people with vestibular dysfunction (Cass et al. 1996)
  - stroke (Bayouk et al. 2006, Bonan et al. 2004)

Cognitive strategy

- ↑ instability and falls during performance of multiple tasks in both neurological and geriatric populations
- Training balance under dual task conditions addresses this.

- **Approach:** Practicing tasks requiring static, anticipatory and reactive balance whilst simultaneously varying the cognitive demands through secondary tasks

- **Secondary Tasks:**
  - Spell words forward/backwards
  - Name any words starting with letters A to K
  - Name any words starting with letters L to Z
  - Remember prices (e.g. bill payment/shopping)
  - Count backward in 3s or 7s or 5s
  - Remember words
  - Visual imaginary task (tell road directions from home to hospital)
  - Recount daily activities

Intervention at the functional level

- Goal: Train ability to adapt balance strategy to changing tasks and environmental demands

- Patient with neurologic damage may have difficulties in perceiving the relevant factors within a specific task or environment

- Practicing balance in a wide variety of tasks and environment is required for optimal recovery





Maximising skills

- Progression
  - Making it more challenging (according to the patient's ability)
  - Complex environments
  - Unpredictable demands
- Approach:
  - Circuit based training
  - Video based exercises
  - Balance Class

- Example of exercises
  - Stepping to pick up objects beyond the person's limit of stability
  - Cross over stepping marks on the floor
  - Lunges in different directions
  - Stepping or dancing in time with music
  - Responding to external timing demands – catching a ball, kicking a ball, computer games
  - Coping with complexity in the environment
    - Negotiating an obstacle course
    - Making sudden stops and starts
    - Stepping over objects
    - Tandem walking
    - Marching on the spot with dual tasking
    - Walking and talking
    - Walking and carrying an object

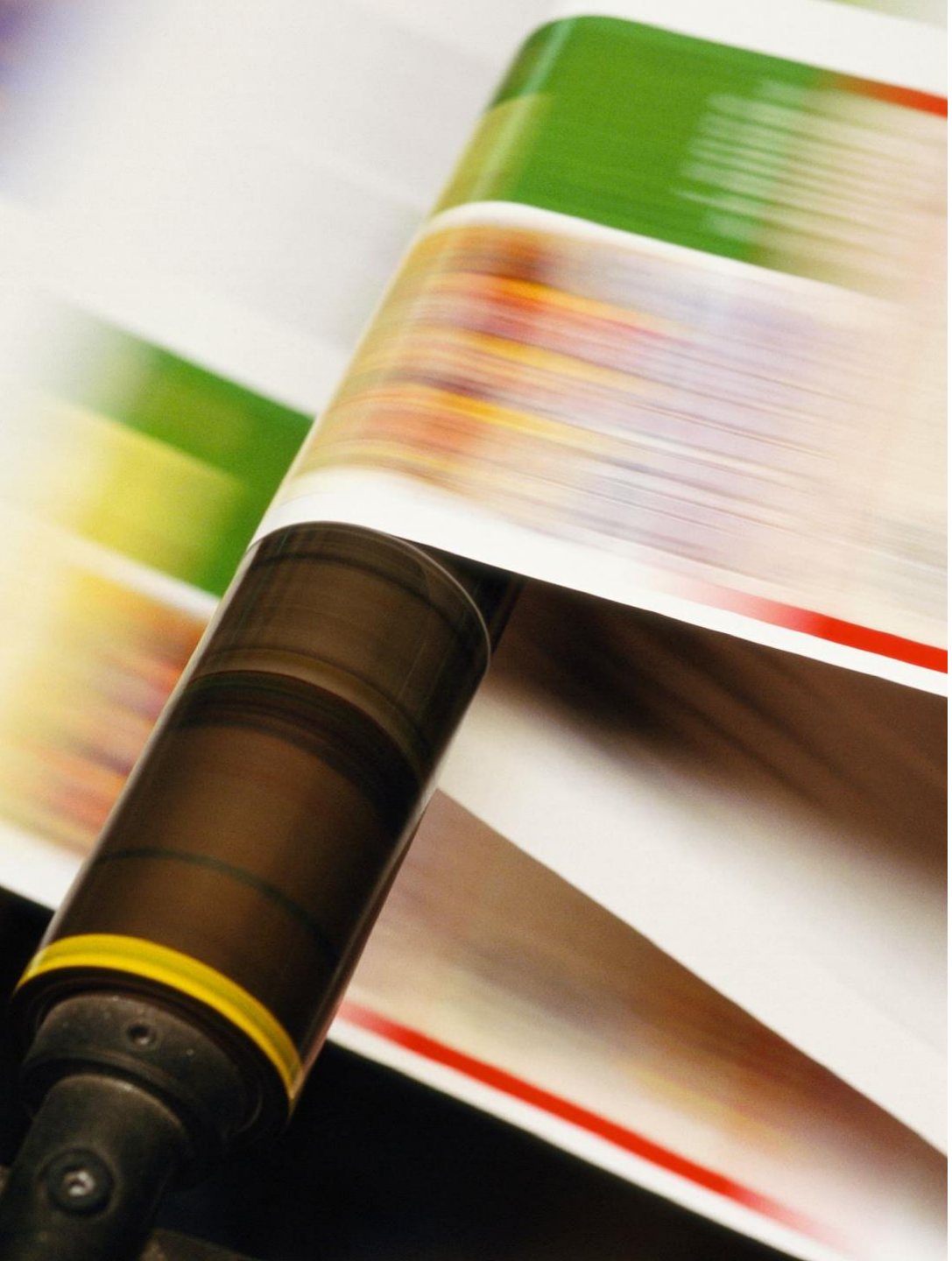
Research evidence for task-oriented balance training

- Van Peppen et al. (2004):https://pubmed.ncbi.nlm.nih.gov/15609840/
  - Systematic review found task-oriented training to restore balance and gait, and for strengthening for the lower paretic limb

- Wevers et al. (2009): https://pubmed.ncbi.nlm.nih.gov/19461035/
  - 6 studies were included and found task-oriented training improved walking distance, gait speed and timed-up and go.

- Cano-Manas et al. (2020): https://pubmed.ncbi.nlm.nih.gov/32148744/
  - Randomised control trial (conventional training + video-based balance training vs conventional training alone, found between group improvements in Tinetti gait assessment, functional reach test, get up and go test.


Any Questions



References

- Shumway-Cook A, Woollacott MH. Motor control: Theory and Practical Applications 4th ed. Philadelphia: Lippincott, Williams, and Wilkins, 2010

- Carr J, Shepherd RB. Neurological Rehabilitation: Optimising motor performance. 2nd Edition. London: Elsevier, Churchill Livingstone, 2010.